

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

IT 19-204-0306 COMPUTER ORGANIZATION AND ARCHITECTURE (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the basic functional units of a computer and its operational concepts.

CO2: Execute a complete instruction and arithmetic operations.

CO3: Understanding of internal organization of memory chips and different memories (static and dynamic).

CO4: Analyse different interrupts, buses and I/O interface circuits.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) A processor has 40 distinct instructions and 24 general purpose registers. A 32 bit instruction word has an opcode, 2 register operands and an immediate operand field. How many number of bits are available for immediate operand field in this instruction format?		3	L3	1	1,5,12
(b) Explain the bus structure of a digital computer.		3	L2	1	1,5,12
(c) Explain the significance of condition code flags in CPU.		3	L2	2	1,3,5,12
(d) Explain about different special purpose registers available inside the CPU.		3	L2	2	1,3,5,12
(e) Why SRAM is costlier than DRAM? Explain with suitable diagrams.		3	L2	3	1,3,5,12
(f) Explain the concept behind cache memory.		3	L2	3	1,3,5,12
(g) Write short notes on Direct Memory Access.		3	L1	4	1,2,3,12
(h) What are the key differences between memory mapped I/O and I/O mapped I/O. In what scenarios in each method typically preferred in computer system design?		3	L4	4	1,2,3,12

PART B

(4 × 12 = 48)

- II. (a) Explain the significance of addressing modes with example. 10 L2 1 1,5,12
- (b) An instruction is stored at location 300 with address field at location 301. The address field has the value 250. What is the effective address, if the addressing mode used is PC-Relative? 2 L3 1 1,5,12

OR

- III. (a) The value of a float type variable is represented using a single precision 32-bit floating point format IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable X is assigned the decimal value of -14.25. What is the representation of X in hexadecimal notation? 5 L3 1 1,5,12
- (b) Using Booth's algorithm perform the multiplication 11 × 14. 7 L3 1 1,5,12

(P.T.O.)

		Marks	BL	CO	PO																														
IV.	Explain the complete execution of the instruction MOV R1,R2 with suitable diagram of internal structure of the CPU.	12	L3	2	1,3,5,12																														
OR																																			
V. (a)	Differentiate between hard-wired and micro-programmed control unit design.	7	L2	2	1,3,5,12																														
(b)	A hard-wired CPU uses 10 control signals S1 to S10 in various time steps T1 to T5 to implement 4 instructions I1 to I4 as shown below. What are the expressions represent the circuits for generating control signals S5 and S10 respectively?	5	L3	2	1,3,5,12																														
<table border="1"> <thead> <tr> <th></th><th>T1</th><th>T2</th><th>T3</th><th>T4</th><th>T5</th></tr> </thead> <tbody> <tr> <td>I1</td><td>S1, S3, S5</td><td>S2, S4, S6</td><td>S1, S7</td><td>S10</td><td>S3, S8</td></tr> <tr> <td>I2</td><td>S1, S3, S5</td><td>S8, S9, S10</td><td>S5, S6, S7</td><td>S6</td><td>S10</td></tr> <tr> <td>I3</td><td>S1, S3, S5</td><td>S7, S8, S10</td><td>S2, S6, S9</td><td>S10</td><td>S1, S3</td></tr> <tr> <td>I4</td><td>S1, S3, S5</td><td>S2, S6, S7</td><td>S5, S10</td><td>S6, S9</td><td>S10</td></tr> </tbody> </table>							T1	T2	T3	T4	T5	I1	S1, S3, S5	S2, S4, S6	S1, S7	S10	S3, S8	I2	S1, S3, S5	S8, S9, S10	S5, S6, S7	S6	S10	I3	S1, S3, S5	S7, S8, S10	S2, S6, S9	S10	S1, S3	I4	S1, S3, S5	S2, S6, S7	S5, S10	S6, S9	S10
	T1	T2	T3	T4	T5																														
I1	S1, S3, S5	S2, S4, S6	S1, S7	S10	S3, S8																														
I2	S1, S3, S5	S8, S9, S10	S5, S6, S7	S6	S10																														
I3	S1, S3, S5	S7, S8, S10	S2, S6, S9	S10	S1, S3																														
I4	S1, S3, S5	S2, S6, S7	S5, S10	S6, S9	S10																														
VI. (a)	How many 128×8 RAM chips are needed to provide a memory capacity of 2048 Bytes? How many lines are there in the address bus? How many of these lines will be common to all chips?	6	L3	3	1,3,5,12																														
(b)	Size of the cache memory is 64 KB. Block size is 32B. Cache memory is mapped with fully associative address mapping. CPU is generating 32 bits of physical address. Cache controller has 1-bit for validity, 1-bit for modification and 2-bits for replacement. Then find: (i) Number of bits for tag field (ii) Size of cache memory controller (iii) Size of cache memory	6	L3	3	1,3,5,12																														
OR																																			
VII. (a)	Consider a fully associative cache with 8 cache blocks numbered 0-7. Consider the following sequence of memory block requests. 4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7 If LRU replacement policy is used, which all blocks will be present in the cache at the end?	6	L3	3	1,3,5,12																														
(b)	Consider a disk with 16 platters, 2 surfaces per platter, 1 K tracks per surface, 2 K sectors per track and 2048 Bytes per sector. Disk rotates with 3000 rpm. Seek time is 10ms. (i) Find capacity of disk. (ii) Find number of bits required for addressing the disk. (iii) Find the disk access time.	6	L3	3	1,3,5,12																														
VIII. (a)	The size of the data count register of a DMAC is 16 bits. The processor needs to transfer a file of 29154 KB from disk to main memory. Here memory is byte addressable. What is the minimum number of times the DMA controller needs to get control of the system bus from the processor to transfer the file from the disk to main memory?	5	L3	4	1,2,3,12																														
(b)	What you mean by interrupts? What are its different types? Explain.	7	L2	4	1,2,3,12																														
OR																																			
IX. (a)	Consider a non-pipelined system which takes 100 ns to perform a task. The same task can be performed using a 6 segment pipeline with cycle time of 19 ns. (i) Calculate the speed up of pipeline for 1000 tasks (ii) What is the maximum speed up?	6	L3	4	1,2,3,12																														
(b)	Explain different pipeline hazards with suitable examples.	6	L2	4	1,2,3,12																														